which shows that $\mathrm{FNN}^{(k(\mathcal{A}),1)}_{\mathcal{A}_i(\mathcal{A})}(d_x, d_y)$ for $i \in [2]$ are universal approximators of $C^0([0,1]^{d_x}, [0,1]^{d_y})$ with respect to the supremum norm.

(4.) All sets of activations that we consider are bounded on any compact set, therefore, Lemma 78 (2.) is applicable. This implies that $\mathrm{FNN}^{(k(\mathcal{A}),1)}_{\mathcal{A}_i(\mathcal{A})}(d_x, d_y)$ are universal approximators of $C^0(\mathbb{R}^{d_x}, \mathbb{R}^{d_y})$ on compact sets with respect to the supremum norm. Since $\mathcal{A} \in \mathcal{S}_1 \cup \mathcal{S}_2$ together with $\mathcal{A}_i(\mathcal{A})$ cover exactly the four sets of activations for which Theorem 11 is formulated, this concludes the proof. ■

Note that the set of activations considered in the previous theorem includes a continuous range of step sizes at zero for the LReLUs. This feature renders standard gradient descent ineffective for training feedforward neural networks with these activations, since variations in the step size do not influence the gradients. The following corollary therefore refines the previous construction to yield a set of LReLUs with only two distinct step sizes, achieved by strategically employing a continuous range of slopes elsewhere in the construction.

Proof of Corollary12. By Lemma 69, the encoder approximation can be constructed via $\mathfrak{e}^\dagger_K \in \mathrm{FNN}^{(d_x)}_{\mathcal{F}^*_{+,1}}(d_x, 1)$. By definition, $\mathcal{F}_+ \subset \mathcal{F}^*_{+,1}$ and $\mathcal{F}_\pm \subset \mathcal{F}^*_{\pm,1}$. Since our memorizer and decoder constructions use only standard LReLUs (without steps), these inclusions simply relax the assumptions on the activation sets. Therefore, we can follow the proof of Theorem 11 to conclude that $\mathrm{FNN}^{(\max\{d_x,d_y\},1)}_{\mathcal{F}^*_{\pm,1}}(d_x, d_y)$ and $\mathrm{FNN}^{(\max\{2,d_x,d_y\},1)}_{\mathcal{F}^*_{+,1}}(d_x, d_y)$ are universal approximators of $C^0(\mathbb{R}^{d_x}, \mathbb{R}^{d_y})$ on compact sets with respect to the supremum norm. ■

Remark 79 *Our coding scheme constructions in Theorems 10, 11, and 18 require a number of layers that grows exponentially with the approximation parameters K and M. Table 3 summarizes the layer counts for each component across different activation classes. We emphasize that these are upper bounds achieved by our constructions; whether more depth-efficient implementations exist at minimal width remains an open question.*

Component	$\mathcal{F}_+, \mathcal{F}_\pm, \mathcal{F}_{+,\mathfrak{s}}, \mathcal{F}_{\pm,\mathfrak{s}}$	**with FLOOR**	$\{\sigma_\alpha\}, \{\sigma_{-\alpha}, \sigma_\alpha\}$
Encoder	$\mathcal{O}(2^K)$	2	$\mathcal{O}(2^K \cdot g_1(\epsilon))$
Memorizer	$\mathcal{O}(2^{d_x K})$	$\mathcal{O}(2^{d_x K})$	$\mathcal{O}(2^{d_x K} \cdot g_2(\epsilon))$
Decoder	$\mathcal{O}(d_y 2^M)$	$\mathcal{O}(d_y 2^M)$	$\mathcal{O}(d_y 2^M \cdot g_3(\epsilon))$

Table 3: Layer counts for our coding scheme constructions. For single-parameter activations, $g_i(\epsilon) \to \infty$ as $\epsilon \to 0$ reflects additional depth from LReLU approximations.

For the activation classes $\mathcal{F}_+$, $\mathcal{F}_\pm$, $\mathcal{F}_{+,\mathfrak{s}}$, and $\mathcal{F}_{\pm,\mathfrak{s}}$, our constructions directly build the required piecewise linear functions:

- *The encoder from Lemmas 67 and 69 approximates the quantizer q_K, which has 2^K linear pieces. By Lemma 61, this requires $\mathcal{O}(2^K)$ layers.*

- *The memorizer from Lemmas 72 and 71 interpolates $2^{d_x K}$ grid points via a piecewise linear function with a comparable number of pieces.*
- *The decoder from Lemma 76 applies min-max strings of length 2^M across d_y output components.*

When FLOOR is available, the encoder can be implemented exactly in depth 2 by Lemma 70, while the memorizer and decoder constructions remain unchanged.
For squashable+FLOOR+Id or STEP+FLOOR+Id activations (Theorem 18), the encoder and decoder have improved depth: the encoder is exact in depth 2 using FLOOR, and the decoder requires only $\mathcal{O}(d_y)$ layers as it is directly constructed from FLOOR and Id activations. However, the memorizer still requires $\mathcal{O}(2^{d_x K})$ layers, which still dominates the overall depth and leads to exponential growth of its depth.
For single-parameter activations $\{\sigma_\alpha\}$ or $\{\sigma_{-\alpha}, \sigma_\alpha\}$, additional layers are needed to approximate the various LReLUs used in our constructions. By Lemma 9, these approximations introduce accuracy-dependent factors $g_i(\epsilon)$ that grow as the target accuracy increases.
In summary, the depth of our constructions scales exponentially (or worse) with the approximation parameters. Since achieving approximation accuracy ϵ typically requires $K, M = \mathcal{O}(\log(1/\epsilon))$, the depth grows polynomially in $1/\epsilon$, which limits direct practical applicability. Our results are primarily of theoretical interest for establishing minimal width universal approximation.

7 Approximating the coding scheme with squashable and FLOOR activations

Lemma 80 *Let $\sigma \in \mathcal{S}$ be a squashable function, then, for all compact intervals $I = [a, b] \subset \mathbb{R}$, $c \in (a, b)$, $\alpha \in \mathbb{R}$, there exists $\phi \in \mathrm{FNN}^{(1)}_{\{\sigma,\mathrm{FLOOR}\}}(1, 1)$ s.t. $\phi|_{[a,b]}(x) = \alpha \cdot \mathrm{STEP}(x - c)$.*

Proof. For $\chi, \epsilon > 0$, by rescaling the width one approximation of step with $1 + 2\epsilon$ from the definition of a squashable functions, there exists a strictly increasing $\psi_{\epsilon,\chi} \in \mathrm{FNN}^{(1)}_{\{\sigma\}}(1, 1)$ s.t. $\|(1 + 2\epsilon)\,\mathrm{STEP} - \psi_{\epsilon,\chi}\|_{I\setminus(-\chi,\chi),\sup} < \epsilon$, $\psi_{\epsilon,\chi}(I) \subset [0, 1 + 2\epsilon]$. Because of the difference in supremum norm, $[\epsilon, 1 + \epsilon] \subset \psi_{\epsilon,\chi}(I)$ and as $\psi_{\epsilon,\chi}$ is strictly increasing, by the mean value theorem there exists $z \in I$ with $\psi_{\epsilon,\chi}(z) = 1$ s.t. for all $x_1 < z < x_2$ it holds that $0 \le \psi_{\epsilon,\chi}(x_1) < 1 = \psi_{\epsilon,\chi}(z)$, $\psi_{\epsilon,\chi}(z) = 1 < \psi_{\epsilon,\chi}(x_2)$. Therefore, we obtain that $\phi(x) := (\alpha\,\mathrm{Id}) \circ \mathrm{FLOOR} \circ \psi_{\epsilon,\chi} \circ (\mathrm{Id}(x) + c - z)$ with $\phi \in \mathrm{FNN}^{(1)}_{\{\sigma,\,\mathrm{FLOOR}\}}(1, 1)$ fulfills $\phi|_I(x) = \alpha \mathbb{1}_{[z,b]}(x + z - c) = \alpha \cdot \mathrm{STEP}(x - c)$. ■

Lemma 81 *Assume that $\mathcal{A}$ is a set of activations with $\mathrm{Id} \in \mathcal{A}$ such that for $\alpha, c \in \mathbb{R}$ it holds that there exists $\psi_{[0,1],\alpha,c} \in \mathrm{FNN}^{(1)}_{\mathcal{A}}(1, 1)$ that fulfills $\psi_{[0,1],\alpha,c}|_{[0,1]}(x) = \alpha\,\mathrm{STEP}(x - c)$. Then, for all $K, M \in \mathbb{N}$, $f \in C^0([0, 1]^{d_x}, [0, 1]^{d_y})$ there exists $\phi \in \mathrm{FNN}^{(3)}_{\mathcal{A}}(1, 1)$ such that*

$$\phi(c) = \mathfrak{m}_{K,M,f}(c) \ \ \forall c \in \mathcal{C}_{d_x K}\ .$$

Proof. We show that for $0 \le x_1 < x_2 < \ldots < x_n$, $y_i \in [0, 1], i \in [n]$ for $n \in \mathbb{N}$ we can construct $\phi \in \mathrm{FNN}^{(3)}_{\mathcal{A}}(1, 1)$ such that $\phi(x_i) = y_i$ for $i \in [n]$, which implies the statement that we want to show. We lead the proof by induction: Set $\phi_0(x) := (1, 1, 1)^T x$ with